

```
> restart;
with(student) :
with(linalg) :
```

```
>
# =====
#      4
#      1
#
#
# =====
```

```
>
# =====
#      4 . 2
#
# =====
```

```
>
assume(L > 0, v > 0);
```

```
>
eq := diff(u(x, t), t$2) / v^2 - diff(u(x, t), x$2) = 0;
```

```
init_c := u(x, 0) = 0, D[2](u)(x, 0) = psi(x) :
psi := x - x :
```

```
bound_c := u(0, t) = 0, u(L, t) = 0;
```

$$eq := \frac{\frac{\partial^2}{\partial t^2} u(x, t)}{v^2} - \frac{\partial^2}{\partial x^2} u(x, t) = 0$$

$$bound_c := u(0, t) = 0, u(L, t) = 0$$

(1)

```
>
# =====
#      : u(x, t) = X(x) * T(t)
# =====
```

```
>
subs(u(x, t) = X(x) * T(t), eq) :
expand(%/(X(x) * T(t))) = 0;
```

$$\left(\frac{\frac{d^2}{dt^2} T(t)}{T(t) v^2} - \frac{\frac{d^2}{dx^2} X(x)}{X(x)} = 0 \right) = 0$$

(2)

```
>
```

```
s1 := diff(T(t), t$2) / (v^2 * T(t)) = -lambda;
s2 := -diff(X(x), x$2) / X(x) = lambda;
```

$$s1 := \frac{\frac{d^2}{dt^2} T(t)}{T(t) v^2} = -\lambda$$

$$s2 := -\frac{\frac{d^2}{dx^2} X(x)}{X(x)} = \lambda$$

(3)

>

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# '      ( )
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```

>

```
assume(lambda > 0);
solX := dsolve(diff(X(x), x$2) + lambda * X(x) = 0, X(x));
X := unapply(rhs(solX), x);
```

$$solX := X(x) = c_1 \sin(\sqrt{\lambda} x) + c_2 \cos(\sqrt{\lambda} x)$$

$$X := x \quad c_1 \cdot \sin(\sqrt{\lambda} \cdot x) + c_2 \cdot \cos(\sqrt{\lambda} \cdot x)$$

(4)

>

```
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#      -
# =====
```

>

```
e1 := X(0) = 0;
e2 := X(L) = 0;
sist := {e1, e2};
```

$$e1 := c_2 = 0$$

$$e2 := c_1 \sin(\sqrt{\lambda} L) + c_2 \cos(\sqrt{\lambda} L) = 0$$

$$sist := \{c_2 = 0, c_1 \sin(\sqrt{\lambda} L) + c_2 \cos(\sqrt{\lambda} L) = 0\}$$

(5)

>

```
A := genmatrix(sist, [_C1, _C2]);
Delta := convert(det(A), trig);
```

$$A := \begin{bmatrix} 0 & 1 \\ \Delta & 0 \end{bmatrix}$$

$$\Delta := -\sin(\sqrt{\lambda} L)$$

(6)

```

>
  _EnvAllSolutions := true;
  solve(Delta = 0, lambda);
  subs(_Zl = k, %);
  ev := unapply(%, k);

      _EnvAllSolutions := true
Warning, solve may be ignoring assumptions on the input variables.

$$\frac{\pi^2 \tilde{Zl}^2}{L^2}$$


$$\frac{\pi^2 k^2}{L^2}$$


$$ev := k \quad \frac{\pi^2 \cdot k^2}{L^2} \tag{7}$$


```

```

>
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#
# =====

```

```

>
  assume(k, posint);
  X := 'X';
  subs(lambda = ev(k), diff(X(x), x$2) + lambda*X(x) = 0) :
  dsolve({%, X(0) = 0, X(L) = 0}, X(x));
  Xsol := rhs(%);
  normX := Xsol / sqrt(int(Xsol^2, x = 0..L));
  normX := simplify(normX, radical, symbolic);
  ef := unapply(normX, (k, x));

      X := X

$$X(x) = c_l \sin\left(\frac{k \tilde{\pi} x}{L}\right)$$


$$Xsol := c_l \sin\left(\frac{k \tilde{\pi} x}{L}\right)$$


$$normX := \frac{c_l \sin\left(\frac{k \tilde{\pi} x}{L}\right) \sqrt{2}}{\sqrt{L c_l^2}}$$


$$normX := \frac{\sin\left(\frac{k \tilde{\pi} x}{L}\right) \sqrt{2}}{\sqrt{L}}$$


$$ef := (k, x) \quad \frac{\sin\left(\frac{k \tilde{\pi} \cdot x}{L}\right) \cdot \sqrt{2}}{\sqrt{L}} \tag{8}$$


```

```
>
# =====
#      (    u ( x , 0 ) = 0 )
# =====
```

```
>
T := 'T';
odeT := diff( T(t), t$2) + v^2*ev(k)*T(t) = 0 :
dsolve( {odeT, T(0) = 0}, T(t));
Tsol := subs(_C1 = 1, rhs(%)) :
Tsol := simplify(Tsol) :
tf := unapply(Tsol, (k, t));
```

$$\begin{aligned}
 T &:= T \\
 odeT &:= \frac{d^2}{dt^2} T(t) + \frac{v^2 \pi^2 k^2 T(t)}{L^2} = 0 \\
 T(t) &= c_1 \sin\left(\frac{v \pi k t}{L}\right) \\
 Tsol &:= \sin\left(\frac{v \pi k t}{L}\right) \\
 Tsol &:= \sin\left(\frac{v \pi k t}{L}\right) \\
 tf &:= (k, t) \quad \sin\left(\frac{v \pi k t}{L}\right)
 \end{aligned} \tag{9}$$

```
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```

```
>
U := (x, t) Sum(C(k)*tf(k, t)*ef(k, x), k = 1 ..infinity);
```

$$U := (x, t) \sum_{k=1}^{\infty} C(k) \cdot tf(k, t) \cdot ef(k, x) \tag{10}$$

```
>
# =====
#      ( )
# =====
```

```
>
eqC := value(subs(t = 0, diff( U(x, t), t))) = psi(x);
```

$$eqC := \sum_{k=1}^{\infty} \frac{C(k) \sqrt{2} \sin\left(\frac{k\pi x}{L}\right)}{L^{3/2}} = x \quad (11)$$

```
# =====
#      '  C ( k )
# =====
```

```
> Ck := Int(psi(x)*ef(k,x),x=0..L)/(v*sqrt(ev(k)));
assume(L > 0):
Ck := value(Ck):
Ck := simplify(Ck) assuming k::posint:
Ck := subs(sin(Pi*k)=0,cos(Pi*k)=(-1)^k,Ck):
Ck := simplify(Ck);
```

$$Ck := \frac{\left(\int_0^L \frac{x \sin\left(\frac{k\pi x}{L}\right) \sqrt{2}}{\sqrt{L}} dx \right) L}{v k \pi}$$

$$Ck := -\frac{\sqrt{2} L^{5/2} (-1)^k}{k^2 \pi^2 v} \quad (12)$$

```
> C := unapply(Ck,k);
```

$$C := k \mapsto -\frac{\sqrt{2} \cdot L^{5/2} \cdot (-1)^k}{k^2 \cdot \pi^2 \cdot v} \quad (13)$$

```
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```

```
> solution := Sum(C(k)*tf(k,t)*ef(k,x),k=1..infinity):
solution;
```

$$\sum_{k=1}^{\infty} \left(-\frac{2 L^2 (-1)^k \sin\left(\frac{v\pi k t}{L}\right) \sin\left(\frac{k\pi x}{L}\right)}{k^2 \pi^2 v} \right) \quad (14)$$

```
> ???
```